
A Forced-Structure Reduction and Verifiable Bounds for Conway’s 99-Graph

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Abstract

Conway’s 99-graph problem asks whether a strongly regular graph with parameters $\text{srg}(99, 14, 1, 2)$ exists. We report a systematic, fully reproducible attack by an autonomous AI research agent, scored under the track’s partial-credit metric. Our verifiable contributions are: (1) an *exhaustive* proof that no circulant graph on $\mathbb{Z}/99$ satisfies more than $3366/4950 = 68.0\%$ of the constraints (33 of 49 difference-classes), with the same ceiling for the other abelian group of order 99; (2) a forced-structure reduction: $\lambda = 1$ makes each neighbourhood a perfect matching and $\mu = 2$ puts the outer vertices in bijection with non-matched neighbour-pairs, collapsing existence to a 12-regular graph on 84 vertices, encoded for CP-SAT and *validated* by recovering the unique $\text{srg}(9, 4, 1, 2)$; (3) a validated prescribed-automorphism orbit-existence framework (fixed-point-free and single-fixed-point actions, checked on $\text{srg}(9, 4, 1, 2)$ and the Paley graph $\text{srg}(13, 6, 2, 3)$), and (4) a best verified artifact at 69.43%, with evidence that this is a robust frontier (fourteen distinct methods, none exceeding it) entangled with the open question, since any provable bound below 4950 is a non-existence proof.

1 Introduction

A *strongly regular graph* $\text{srg}(v, k, \lambda, \mu)$ is a k -regular graph on v vertices in which every adjacent pair has exactly λ common neighbours and every non-adjacent pair has exactly μ . The parameter set $(99, 14, 1, 2)$ was identified by Biggs [3] and popularised by Conway, who offered a \$1000 prize for a construction or a non-existence proof [18, 7]. The parameters pass every standard feasibility test (integral eigenvalues $14^{(1)}, 3^{(54)}, (-4)^{(44)}$, the Krein conditions, the absolute bound [5, 4]), and it is known that such a graph cannot be vertex-transitive, which eliminates all standard algebraic constructions [6]. The problem is open.

Modern SAT, SMT, and CP-SAT solvers [8, 2, 14] have been applied across combinatorial and synthesis problems [9, 11, 15–17]; they are the main tools used below.

We attempt the `conways-99-graph` instance of the CAISc 2026 Verifiable Problems track [13]. The track scores a submitted 99×99 symmetric 0/1 matrix by the fraction of constraints satisfied: 99 degree constraints (row sums 14), one λ constraint per edge, and one μ constraint per non-edge. As every unordered pair is an edge xor a non-edge, the denominator is fixed at $99 + \binom{99}{2} = 4950$, and a perfect score is a solution to the open problem. We attack the well-posed sub-question: *how high can the score be pushed, and what can be proved about it?* Per the track’s stated interest, we report how an AI agent navigated the problem. We are explicit that we neither construct the graph nor prove non-existence; our contributions are verifiable bounds, a validated reduction, a reusable framework, and the documented trajectory.

2 Problem, metric, and a non-obstruction

Let A be the adjacency matrix and $C = A^2$, so C_{ij} counts common neighbours of i, j . The strongly-regular conditions are equivalent to the single matrix identity $A^2 + A - 12I = 2J$ (with J all-ones); the score counts how many of its 4950 scalar instances hold.

For any 14-regular graph, $\sum_{i < j} C_{ij} = \sum_v \binom{\deg v}{2} = 99 \binom{14}{2} = 9009$, while the targets sum to $693 \cdot 1 + 4158 \cdot 2 = 9009$: *exactly equal*. The feasible spectrum fixes $\sum_{i < j} C_{ij}^2 = 17325$, again matched by the targets. Thus first and second moments give no contradiction, which is precisely why the problem resists easy arguments. A practical corollary, which shapes every search below, is that near degree 14 the λ - and μ -satisfaction are *coupled*: one cannot cheaply maximise μ alone, because the common-neighbour budget is tight, so a high partial score forces genuine near-strong-regularity.

3 An exhaustive bound for circulant graphs

A circulant on $\mathbb{Z}/99$ is given by a symmetric connection set $S = -S$ with $|S| = 14$; it is automatically 14-regular. By vertex-transitivity, all pairs in a difference-class $\{d, -d\}$ behave identically, so the score is $99 + 99 \cdot (\text{satisfied difference-classes})$, over 49 classes, where class d is satisfied iff ($d \in S$ and the cyclic autocorrelation $\lambda(d) = |S \cap (S + d)| = 1$) or ($d \notin S$ and $\lambda(d) = 2$). A perfect circulant would be a $(99, 14, 1, 2)$ partial difference set, known not to exist [12]; we compute the best partial one.

Proposition 1 (Exhaustive) *Over all $\binom{49}{7} = 85,900,584$ symmetric connection sets, the maximum number of satisfied difference-classes is 33; the best circulant scores $99 + 99 \cdot 33 = 3366/4950 = 68.0\%$.*

We verified Proposition 1 by complete enumeration with batched FFT autocorrelation (≈ 100 s on a laptop). An optimal set is $S = \{\pm 1, \pm 2, \pm 4, \pm 15, \pm 27, \pm 36, \pm 45\}$, with degree 99/99, λ 198/693, μ 3069/4158; the algebra concentrates non-edge common-neighbour counts at 2 (μ at 73.8%, versus $\approx 27\%$ for a random regular graph). The non-cyclic abelian group of order 99, $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_{11}$, attains the same 33/49.

4 A forced-structure reduction, validated on $\text{srg}(9, 4, 1, 2)$

Most of any $\text{srg}(v, k, 1, 2)$ is forced. Fix a vertex 0 with $N(0) = \{1, \dots, k\}$. Since $\lambda = 1$, each neighbour shares exactly one common neighbour with 0, so the first subconstituent $N(0)$ is a perfect matching [5]; fix it (WLOG) as $(1, 2), (3, 4), \dots$. Since $\mu = 2$ on non-edges $(0, \text{outer})$, every outer vertex has exactly two neighbours in $N(0)$; and since $\mu = 2$ on inner non-edges and $\lambda = 1$ on inner edges, the outer vertices are in bijection with the non-matched pairs of $N(0)$. Hence the inner–outer adjacency is *entirely forced*: the outer vertex labelled $\{a, b\}$ is adjacent to exactly inner a and b .

The sole unknown is the outer–outer graph: $(k-2)$ -regular on $M = \binom{k}{2} - k/2$ vertices, constrained by label-derived λ/μ conditions. With inner vertices $0, \dots, k-1$ and matching $m(2i) = 2i+1$, the outer vertices are the non-matched pairs. For outer u, v with labels P_u, P_v and $s = |P_u \cap P_v| \in \{0, 1\}$: if $u \sim v$ then the number of common outer neighbours equals $1 - s$, else $2 - s$. For inner a and outer u : if $a \in P_u$, exactly one outer neighbour of u contains a ; if $a \notin P_u$, then $|m(a) \cap P_u|$ plus the number of outer neighbours of u containing a equals 2. The symmetry generators are the $k/2$ within-pair swaps and the $k/2 - 1$ adjacent-pair transpositions; each induces a permutation of the outer vertices (hence of the edge variables), to which a lex-leader constraint is applied.

For $(99, 14, 1, 2)$ this is a 12-regular graph on 84 vertices.

We encode the reduced problem for CP-SAT and add lex-leader symmetry breaking over the group $B_{k/2}$ that relabels the $k/2$ matching-pairs and swaps within them (order $2^7 \cdot 7! = 645,120$ for $k = 14$), via its 13 generators. The pipeline is *validated end-to-end*: it recovers the unique $\text{srg}(9, 4, 1, 2)$ in milliseconds, with and without symmetry breaking. The $(99, 14, 1, 2)$ model has 379,987 Booleans and 761,221 constraints; in our runs it neither returns a graph nor exhausts (the expected outcome for an open problem), so we release it as a validated, maximally-pruned framework.

Table 1: Representative results across the fourteen configurations we ran (rows group related methods); the score converges to 68.0–69.43%. The best verified artifact is $3437/4950 = 69.43\%$.

Method	character	best score
Exhaustive circulant ($\mathbb{Z}/99$)	proven bound	68.0%
Cayley search ($\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_{11}$)	search	68.0%
Block CP-SAT, $\mathbb{Z}/11, \mathbb{Z}/9, \mathbb{Z}/3$	exact, symmetric	68.0%
Full MaxSAT (504,504 vars)	exact	68.0%
Degree-preserving 2-opt SA	heuristic	$\approx 56\%$
Tabu / ILS	heuristic	69.3%
Min-conflicts; blended SA + island EA	heuristic	69.43%

5 Prescribed-automorphism search and a negative methods finding

The research frontier is the existence question under a prescribed automorphism. Cesarz and Woldar [6], building on the orbit-matrix method of Behbahani and Lam [1], prove that Aut of any such graph is severely constrained: orders 9 and 11 are *excluded*; if $2 \mid |G|$ then $|G| \mid 6$; and if $7 \mid |G|$ then $G \cong \mathbb{Z}_7$. Order 7 is thus constrained but *not* ruled out: whether a $\mathbb{Z}_7$ -symmetric $\text{srg}(99, 14, 1, 2)$ exists is itself open. Since the non-fixed vertices split into 7-cycles, any order-7 action fixes $f \equiv 99 \equiv 1 \pmod{7}$ vertices; the minimal case has 14 orbits of size 7 and a single fixed point.

We built a clean orbit-model existence encoding for a prescribed automorphism of order p . With $B = 99/p$ orbits, an invariant graph is block-circulant (adjacency of $(i, a), (j, b)$ depends only on $(i, j, b - a \bmod p)$), and for every pair-class the strong-regularity collapses to *common* + *adjacency* = 2, encoding $\lambda=1$ and $\mu=2$ at once. We support both the fixed-point-free case (semiregular $p \mid 99$) and the single-fixed-point case (the fixed vertex joins full orbits, forced by $14 = 2 \cdot 7$ to exactly two of them). The encoding is *validated end-to-end*: it reconstructs and re-verifies $\text{srg}(9, 4, 1, 2)$ (both a fixed-point-free $\mathbb{Z}_3$ and an order-2 action with one fixed point) and the Paley graph $\text{srg}(13, 6, 2, 3)$ (order-3, one fixed point).

We then attacked the genuinely open sub-cases. For the single-fixed-point $\mathbb{Z}_7$ model (the minimal admissible order-7 action, where a construction would resolve existence and an infeasibility certificate would eliminate it), CP-SAT returns UNKNOWN even after a 48-hour run on 14 cores, neither building the graph nor proving infeasibility; the fixed-point-free $\mathbb{Z}_3$ model (33 orbits) likewise returns UNKNOWN within 1800 s. We record this as an honest negative methods finding: even on an open sub-case where a specialised orbit-matrix enumeration would terminate, off-the-shelf CP-SAT does not, in our hands, decide the instance, and its persistence across a $96\times$ longer budget points to a structural barrier in the general-purpose encoding rather than a mere time shortfall. This sharply delimits the general-purpose approach and motivates the specialised orbit-matrix + eigenvalue-interlacing machinery.

6 Heuristic frontier, best artifact, and search trajectory

For general (asymmetric) graphs we built an $O(\deg)$ incremental engine that maintains $A, C = A^2$, the squared error $\text{SE} = \sum_{i < j} (t_{ij} - C_{ij})^2$ (target $t_{ij} = 1$ if edge else 2; a fitness used in a prior evolutionary-algorithm attempt [10]), and the exact-match score, so an edge toggle updates all in $\approx O(14)$ time. Because exact-match is a flat objective (no gradient), local search stalls on it; we instead optimise the *blend* $O(A) = \text{real}(A) - \alpha \text{SE}(A)$, which keeps the true objective primary while $-\alpha \text{SE}$ supplies a descent direction across plateaus, inside an island-model evolutionary algorithm with degree-preserving crossover.

Table 1 summarises the frontier; the best artifact has degree $69/99$, λ 374/708, μ 2994/4143. Every high-scoring solution sits near $\lambda \approx 53\%$, $\mu \approx 72\%$, the circulant is a strict 2-opt local maximum, and large-neighbourhood CP-SAT re-optimisation of 14-vertex chunks yields only lateral moves. Restarting min-conflicts *from* the best artifact with elevated noise explored 1.57×10^6 accepted moves without satisfying a single additional constraint. The frontier is a strict local optimum, not a tuning artifact. To our knowledge, no prior work reports a partial-credit score on the track’s constraint

metric; the prior evolutionary attempt of Hutnyk [10] optimised squared error, not the constraint count.

Agent trajectory (reported per the track’s interest). The agent first encoded the problem for an SMT solver and stalled on the global connectivity/structure constraints; a corrective pivot to a CP-SAT solver gave order-of-magnitude speedups and unlocked the exhaustive circulant bound and the block models. Pushed for rigour, the agent then derived and validated the forced-structure reduction (Section 4), implemented the orbit-existence encoding, and, after a literature review establishing the open status and the automorphism results, re-scoped its claims to verifiable bounds and an explicit non-claim on existence. A final construction attempt prioritised the single most structurally-justified model (the open single-fixed-point $\mathbb{Z}_7$ orbit case) over undirected search, ran it to a 48-hour budget, and reported its inconclusive (UNKNOWN) outcome as such. The repeated theme was course-correction away from an over-optimistic framing toward certifiable statements.

7 Discussion and limitations

A score of 4950 is an $\text{srg}(99, 14, 1, 2)$; hence any provable upper bound below 4950 would be a non-existence proof, and any method reaching 4950 an existence proof. Crossing the partial-credit frontier toward the “high 90s” is thus not a separate engineering target but is entangled with the open problem: a 95% near-SRG is as structurally delicate to find as the graph. **Limitations.** We do not resolve existence; our 69.43% artifact is a partial score, the $\approx 69\%$ frontier is a search-landscape observation (not a proven global bound), and general-purpose CP-SAT did not decide the prescribed-automorphism sub-cases, including the genuinely open single-fixed-point $\mathbb{Z}_7$ case, which stayed UNKNOWN even after a 48-hour, 14-core run. The circulant bound (Prop. 1) and the reduction validation are the rigorous, reproducible results.

8 Conclusion

We contribute an exhaustive circulant ceiling (68.0%), a validated forced-structure reduction to a 12-regular graph on 84 vertices, a validated orbit-existence framework with an honest negative finding on the open single-fixed-point $\mathbb{Z}_7$ sub-case, a best verified artifact at 69.43%, and a documented AI-agent trajectory, offered (as the track frames search work) as reusable infrastructure and structural pruning, with no claim on the open existence question.

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Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

For items 1-4, give a score from the scale below that defines the role of AI in each part of the scientific process. The scores are as follows:

- **[A] Human-generated:** Humans generated 95% or more of the research, with AI being of minimal involvement.
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For each research stage where AI was involved, i.e. where you selected **[B]**, **[C]**, or **[D]**, please also indicate the approximate level of iteration effort required using the provided iteration macros. Count substantive attempts, prompts, runs, agent trajectories, or course corrections; do not count minor wording edits.

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These categories leave room for interpretation, so we ask that the authors also include a brief explanation elaborating on how AI was involved in the tasks for each category. Please keep your explanation to less than 150 words.

1. **Hypothesis development:** Hypothesis development includes the process by which you came to explore this research topic and research question. This can involve the background research performed by either researchers or by AI. This can also involve whether the idea was proposed by researchers or by AI.

Answer: **[B]**

Iteration Effort: **[Medium]**

Explanation: The problem instance and the strategic plan (the approaches, use of constraint solvers, considering the $\mathbb{Z}/9 \times \mathbb{Z}/11$ structure, and consult the literature) came from the author. The implementation and refinement of approaches (the circulant autocorrelation bound, the forced-structure reduction, the orbit-existence and blended-objective formulations) were generated by a frontier AI agent across iterations.

2. **Experimental design and implementation:** This category includes design of experiments that are used to test the hypotheses, coding and implementation of computational methods, and the execution of these experiments.

Answer: **[C]**

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Explanation: The agent wrote the code (solver encodings, the reduction and its validation, and the fixed-point-free and single-fixed-point orbit-existence models).

3. **Analysis of data and interpretation of results:** This category encompasses any process to organize and process data for the experiments in the paper. It also includes interpretations of the results of the study.

Answer: **[B]**

Iteration Effort: **[Low]**

Explanation: The results were primarily analysed by the author, and the AI agent was only used for the final polish (to check if anything is missing).

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Answer: [C]

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Explanation: The AI agent drafted the manuscript and tables. The author directed reviewing, revisions of framing, and scope.

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For items 5-6, provide a free-text description.

5. **AI systems used:** Describe all AI systems used in this research without naming specific products or models. Use system-level descriptors only (e.g., “a frontier large language model”, “a protein structure prediction system”, “a custom multi-agent framework built on open-source models”). Include relevant details such as system type, scale, and capabilities. **Specific model and product names should only be included in the camera-ready version.**

Description: Anthropic’s Claude, run as an autonomous research-and-coding agent via the Claude Code CLI with shell and tool access, invoking external constraint solvers (Google OR-Tools CP-SAT and Microsoft Z3) and standard scientific-Python libraries (NumPy, SciPy) for enumeration and verification.

6. **Observed AI Limitations:** What specific limitations and failure modes have you observed when using AI as a partner or lead author?

Description: (i) recurring tendency to over-claim or under-scope results, requiring repeated correction toward certifiable statements; (ii) inability of its general-purpose solver to decide the prescribed-automorphism instances (including the single-fixed-point $\mathbb{Z}_7$ sub-case).

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Answer: [Yes]

Justification: The abstract and introduction state exactly the verifiable contributions (exhaustive circulant bound, validated reduction, orbit framework with a negative finding, 69.43% artifact) and explicitly disclaim any resolution of existence.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Proposition 1 is a computational claim proved by complete enumeration; the assumptions (symmetric $|S| = 14$ connection set on $\mathbb{Z}/99$; satisfaction criterion per difference-class) are stated in-place, and the enumerator and verifier are released with the paper. The forced-structure reduction of Section 4 states its assumptions ($\lambda = 1, \mu = 2$) and derives the inner matching, inner-outer bijection, and outer-graph constraints in the text.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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Justification: The circulant enumeration (search space, autocorrelation criterion, running time), the reduced CP-SAT model (379,987 Booleans, 761,221 constraints, symmetry-breaking generators for $B_{k/2}$), the orbit-existence encoding, the blended objective ($O(A) = \text{real}(A) - \alpha \text{SE}(A)$), and the compute budgets are all specified in the main text; source and the 69.43% artifact are released as supplementary material.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Answer: [Yes]

Justification: All experiments ran on a single 14-core laptop CPU; the exhaustive circulant enumeration takes ≈ 100 s, the prescribed-automorphism orbit searches were run for 1800 s (the open single-fixed-point $\mathbb{Z}_7$ case additionally to a 48-hour budget), and other solver runs take minutes to a few hours, as stated.

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- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations, privacy considerations, and security considerations.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies.